WJEC CBAC

GCE MARKING SCHEME

Predicted Paper (Hard)

ECONOMICS – UNIT 3

1520U30-1

About this marking scheme

This marking scheme follows the WJEC positive marking principles: credit what the learner writes rather than penalising omissions.

Learners must demonstrate ability to:

- AO1 — Demonstrate knowledge of terms/concepts and theories/models.
- AO2 — Apply knowledge to various economic contexts.
- AO3 — Analyse issues, showing understanding of impact on economic agents.
- AO4 — Evaluate arguments using qualitative and quantitative evidence.

GCE A LEVEL ECONOMICS – UNIT 3

PREDICTED PAPER (HARD) MARK SCHEME

Question 1 (a) — Internal economies vs diseconomies of scale (4 marks)

AO1 – 2 marks

- 1 mark for understanding: economies reduce average cost as output rises; diseconomies raise it.
- 1 mark for accurately drawn LRAC curve (U-shaped, correctly labelled). Do not accept plain 'AC'.

AO2 – 2 marks

- 1 mark for a Toyota internal economy of scale: robotic assembly at Tsutsumi, long production runs, Toyota Production System reducing unit costs.
- 1 mark for a Toyota internal diseconomy: 2010 recall / communication failures between divisions; bloated middle-management layers; production errors at overseas plants.

Question 1 (b) — External economies of scale in Aichi (4 marks)

AO1 – 2 marks

- 1 mark for understanding that external EoS reduce LRAC at every level due to growth/changes outside the firm but within the wider industry/area.
- 1 mark for correct diagram showing LRAC shifting downward to $LRAC_1$.

AO2 – 2 marks

- 1 mark for linking to case: keiretsu supplier network, local engineering schools, subsidised transport infrastructure in Aichi.
- 1 mark for linking data to falling LRAC: a dense supplier base reduces just-in-time delivery costs; local engineering graduates reduce recruitment and training costs; better transport reduces logistics/congestion costs.

Question 2 (a)(b)(c) — Kinked demand oligopoly diagram (7 marks)

(a) 1 mark (AO2): output indicated at the level where MC = MR (i.e. output at the kink point of AR, directly below the discontinuity in MR).

(b) 2 marks (AO2): total fixed costs shown as the rectangle between AC and AVC at the firm's output (i.e. AFC × Q). Candidates may use AFC curve if shown; OFR from (a) allowed.

Note: if AVC curve not shown, candidates may demonstrate TFC via AFC × Q × height from AFC curve.

(c) 4 marks: price stability under kinked demand.

- AO1 (1 mark): knowledge that MR has a vertical discontinuity where AR kinks.
- AO2 (1 mark): MC may shift within this discontinuity without altering MR = MC output.
- AO3 (2 marks): developed chain — above P*, rivals are assumed not to follow a price rise, so AR is elastic and TR falls; below P*, rivals match price cuts, so AR is inelastic and TR also falls. Either way the firm loses from changing price, so the profit-maximising price remains stable at P* despite moderate MC changes.

Question 3 — Evaluate Japan Post privatisation (8 marks)

Band	AO2 (2)	AO3 (2)	AO4 (4)
3			**4 marks — Excellent evaluation, judgement weighing both sides well.**
2	**2 marks — Data used well on both sides.**	**2 marks — Good detailed analysis with developed chains of reasoning.**	2–3 marks — Well-developed counter-argument(s).
1	**1 mark — Superficial use of data or on one side only.**	**1 mark — Limited/superficial analysis.**	**1 mark — Counter-argument present but not developed.**
0	0	0	0

In favour of success:

- Raised capital for the Japanese government, relieving pressure on public finances.
- Exposed Japan Post to market discipline — potential for greater efficiency.
- Capital previously locked into JGBs can now be allocated more productively.
- Remains a major player in Japan's financial system (¥200tn assets).
- Competition with private banks/insurers theoretically drives product innovation.

Against success:

- ¥400bn Toll Holdings write-down suggests poor strategic decision-making post-privatisation.
- Share price underperformance suggests market sees limited upside.
- 2021 mis-selling scandal at Japan Post Insurance shows governance failures.
- Service quality concerns for rural/elderly customers.
- Government still holds a majority stake — limited genuine privatisation; principal-agent issues persist.
- Privatisation does not address Japan's deeper deflationary savings culture that channels funds through postal accounts.

Reversible answer.

Question 4 — Negative interest rates and AD in Japan (6 marks)

Band	AO1 (2)	AO2 (2)	AO3 (2)
2	**2 marks — Good knowledge of how negative rates affect savers AND borrowers.**	**2 marks — Good use of data (specific reference to rate of −0.10% from 2016 to March 2024, shift to +0.25% by July 2024).**	**2 marks — Good analysis of specific AD components.**
1	**1 mark — Knowledge limited to one side.**	**1 mark — At least one data reference.**	**1 mark — Superficial analysis of AD as a whole.**
0	0	0	0

Indicative content:

- Negative rates impose a cost on commercial bank reserves held at the BoJ, incentivising banks to lend.
- C: Japanese households famously save a high share of income; negative rates aim to encourage spending, though cultural factors limit effect.
- I: cheaper borrowing should raise investment, but Japanese firms held large cash reserves, limiting the boost.
- G: low/negative rates reduce debt-service costs on Japan's 263%-of-GDP public debt, sustaining G.
- X–M: hot-money outflow weakens the yen (from ¥115 to ¥150 by Oct 2022), boosting export competitiveness and inbound tourism — rising X.
- However, with inflation still low for most of the period, policy transmission was weak — a 'liquidity trap' dynamic.

Question 5 (a) — Japan price index calculation (3 marks)

Weighted method:

(14.6 × 7) + (9.0 × 22) + (7.4 × 12) + (1.8 × 50) + (0.4 × 7) + (3.7 × 2) / 100

= (102.2 + 198 + 88.8 + 90 + 2.8 + 7.4) / 100 = 489.2 / 100 = 4.892%

Price index = 104.9 (accept 104.8 – 104.9)

- AO3 (1 mark) for use of weighted index method.
- AO2 (2 marks) for correct answer.
- 2 marks in total for 4.89% without expressing as an index.

Question 5 (b) — Assess wage-price spiral in Japan (8 marks)

Band	AO2 (2)	AO3 (2)	AO4 (4)
3			**4 marks — Excellent evaluation weighing both sides.**
2	**2 marks — Good use of data with references to at least two Figures.**	**2 marks — Good analysis of spiral mechanism.**	2–3 marks — Well-developed counter-argument.
1	**1 mark — Data on one side only.**	**1 mark — Superficial.**	**1 mark — Counter not developed.**
0	0	0	0

In favour of a spiral:

- Fig 1 shows inflation rising from −0.2% (2021) to 3.1% (2023) — first sustained rise in decades.
- Fig 2 shows essentials up sharply (energy 14.6%, food 9.0%) — categories with low demand elasticity feeding into expectations.
- 2023 shuntō delivered 3.6% wage rises, 2024 shuntō 5.1% — largest in 30 years, breaking Japan's deflationary wage psychology.
- Firms beginning to pass through prices (Yamazaki, Kewpie) — previously rare.

- Weak yen keeps imported inflation high, sustaining pay pressure.

Against a spiral:

- Fig 3 shows real wages fell from 104 (2019) to 100 (2023) and have been negative for 21 months — workers have lost ground, not gained it.
- Japan's decades of deflation mean inflation expectations remain anchored near zero.
- BoJ has begun tightening (rate to 0.25% by July 2024) with further rises signalled — anchors expectations.
- Services inflation only 1.8% — services are labour-intensive so would be the first sign of a spiral. Low services inflation suggests wage pressure is not translating to widespread price rises yet.
- Ageing population and rigid labour market limit worker bargaining power economy-wide (headline shuntō numbers apply mainly to large firms).
- Global commodity prices are stabilising, so imported cost-push is easing.

Reversible answer.

Section B — Japan's yen crisis and Abenomics reckoning

Question 6 (a)(i) — % change in yen LNG price (2 marks)

(143,000 − 47,000) / 47,000 × 100 = 96,000 / 47,000 × 100 ≈ 204.3%

Accept 204%, 204.2%, 204.3%.

Question 6 (a)(ii) — Yen price of imported oil, constant USD price (2 marks)

Yen weakened from ¥103/US$ to ¥150/US$ — that is, ¥150/¥103 = 1.456 times more yen per dollar (1 mark for interpreting the FX change).

If the US$ price of oil was constant, the yen price of oil rose by (1.456 − 1) × 100 = 45.6% (1 mark).

Accept 45% – 46%.

Alternative method: % change = (150 − 103) / 103 × 100 = 45.6%.

Question 6 (b) — Cost & revenue diagram for Japanese importing firms (6 marks)

Band	AO1 (4)	AO2 (2)
3	**4 marks — Accurate diagram: higher MC and AC, higher price, lower output, lower profit.**	
2	2–3 marks — Largely accurate with one labelling error (3 marks) or multiple errors (2 marks).	**2 marks — Data convincingly used.**
1	**1 mark — Weak diagram with multiple errors.**	**1 mark — Data referenced.**
0	0	0

Indicative AO1:

- MC and AC shift up (MC_2, AC_2).
- New lower output at MR = MC_2.
- New higher price P_2.
- Lower abnormal profit area.

AO2:

- LNG up 204%, crude oil up 168% in yen terms (Table 1).
- Yen weakened ~45% vs US$ between 2021 and Oct 2022, amplifying imported cost rises.
- Case states 30% YoY rise in corporate bankruptcies in 2023 — supports 'falling profits' part of the question.

Question 7 — External vs internal factors in Japan's 2022–24 difficulties (8 marks)

Band	AO2 (2)	AO3 (2)	AO4 (4)

3			**4 marks — Well-rounded judgement recognising both. Top band sees problems as a mix.**
2	**2 marks — Data used well on both sides.**	**2 marks — Well-developed analysis.**	2–3 marks — Well-developed counter-arguments.
1	**1 mark — Data used superficially.**	**1 mark — Limited chains of reasoning.**	**1 mark — Limited evaluation.**
0	0	0	0

External factors:

- Global energy/food price spikes from 2021 onward due to pandemic recovery and the war in Ukraine.
- US Federal Reserve raising rates aggressively widened the US-Japan interest-rate differential, driving yen depreciation and 'carry-trade' outflows — largely beyond BoJ control.
- Japan's 90% oil and 60% food import dependence makes it particularly exposed to global price shocks.
- World commodity prices priced in US$ mean yen weakness magnified imported inflation.

Internal factors:

- BoJ's persistent ultra-loose policy (−0.10% through 2022) widened the interest-rate gap vs the US, weakening the yen — a domestic policy choice.
- Abe's 'third arrow' of structural reform largely undelivered (case reference).
- Ageing population (old-age dependency 50.3% — Table 2) and shrinking workforce limit growth.
- Low productivity growth (0.5% p.a. since 2000) limits real wage growth and tax base.
- Labour market rigidities and corporate governance weaknesses predate the crisis.
- 263% public debt/GDP constrains fiscal response — a legacy of internal policy choices.

Reversible answer.

Question 8 — Should the BoJ have intervened to strengthen the yen? (10 marks)

Band	AO2 (4)	AO3 (3)	AO4 (3)
3	**4 marks — Data embedded, both intervention types covered on both sides.**	**3 marks — Excellent analysis of both rate rises AND FX intervention.**	**3 marks — Well-rounded judgement covering both forms.**
2	2–3 marks — Data used well, may focus on one intervention.	**2 marks — Well-developed analysis, narrower range.**	**2 marks — Well-developed counter-argument.**
1	**1 mark — Limited data use.**	**1 mark — Limited analysis.**	**1 mark — Limited evaluation.**
0	0	0	0

Support intervention:

- Weak yen is driving imported inflation: LNG up 204%, oil up 168% in yen terms — hitting real incomes.
- Stronger yen would cut imported commodity prices and help anchor inflation.
- Japanese consumer protest at repeated price rises (Yamazaki, Kewpie) indicates political pressure.
- ¥9tn intervention in Sep/Oct 2022 did temporarily stabilise the yen — evidence that large-scale intervention works short-term.
- Corporate bankruptcies up 30% YoY in 2023 — damage to import-dependent firms justifies action.
- Rate rises in July 2024 (to 0.25%) did eventually help stabilise yen.

Against intervention:

- Interest rate hikes would raise debt-service costs on 263%-of-GDP public debt — fiscal damage potentially greater than FX benefit.
- Carry-trade unwind risk — a rapid BoJ hike could trigger global market volatility (as seen briefly in August 2024).
- Rate rises risk tipping Japan back into deflation — undoing a decade of reflation effort.
- FX intervention is costly: Japan's reserves fell from $1,386bn (Jan 2022) to $1,194bn (Oct 2022), a drop of $192bn — and the trend was not reversed.
- No central bank can sustainably fight a structural trend (yen weakness driven by US-Japan rate gap) — intervention wastes reserves.
- Weak yen benefits Japan's export champions (Toyota, Sony) and inbound tourism — do you really want to strengthen the yen?

If only one form of intervention is evaluated, maximum Band 2 (7/10). Reversible answer.

Question 9 — Will structural reform improve Japanese living standards? (12 marks)

Band	AO2 (4)	AO3 (3)	AO4 (5)
3	**4 marks — Case embedded throughout on both sides.**	**3 marks — Excellent analysis focused on living standards indicators.**	**5 marks — Well-rounded judgement.**
2	2–3 marks — Case well used but uneven coverage.	**2 marks — Developed analysis, narrower range or less LS focus.**	3–4 marks — Well-developed counter-arguments.
1	**1 mark — Superficial case use.**	**1 mark — Limited analysis.**	1–2 marks — Limited evaluation.
0	0	0	0

In favour of improved living standards:

- Greater immigration tackles old-age dependency (50.3% — highest in G7, Table 2), supports the tax base and pension system.
- Higher female workforce participation (already 73.6% — Table 2) could rise further if childcare and promotion pathways reform.

- Corporate governance reform (unlocking ¥500tn in corporate cash reserves) could fund productivity-raising investment, directly raising real wages.
- Diversifying the energy mix would reduce LNG import dependence (LNG price up 204% — Table 1), insulating future real incomes.
- Rising productivity (currently 53.4 vs G7 71.8) would enable sustainable real wage growth.
- A lower old-age dependency ratio reduces pressure on healthcare and pension spending.

Against:

- Immigration remains politically sensitive in Japan; previous reforms have been modest.
- Abenomics 'third arrow' failed previously — institutional inertia is substantial.
- Short-term adjustment costs: deregulation displaces incumbent workers, hurting LS before gains materialise.
- Public debt at 263% of GDP (Table 2) constrains supportive fiscal spending for transition costs.
- Structural reform alone does not address short-term cost-of-living hit from weak yen and imported inflation.
- LS already high in some dimensions (Japan's life expectancy and low crime); marginal improvements may be harder to achieve than the case implies.
- Cultural resistance to 'lifetime employment' reform may slow benefits.

Band 3 requires a clear focus on living-standard indicators and the impact of reform on them.

AO Grid

Section A

Question	AO1	AO2	AO3	AO4	Total
1	4	4	–	–	8
2	1	4	2	–	7
3	–	2	2	4	8
4	2	2	2	–	6
5	–	4	3	4	11
Total	7	16	9	8	40

Section B

Question	AO1	AO2	AO3	AO4	Total
6(a)	–	4	–	–	4
6(b)	4	2	–	–	6
7	–	2	2	4	8
8	–	4	3	3	10
9	–	4	3	5	12
Total	4	16	8	12	40